

Markscheme

Mathematics: analysis and approaches
Practice question bank

93. (a)

It can be shown that $(n + 1)^2 - n^2 = 2n + 1$ for every integer n , by expanding the left-hand side.

Using this result, find the value of n for which $(n + 1)^2 - n^2 = 57$.

$$(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1 \quad (M1) \text{ A1}$$

$$\text{set } 2n + 1 = 57 \quad (M1)$$

$$n = 28 \quad \text{A1}$$

[4 marks]